

CAPE MAY, NJ, USA

WMO: 745966

Lat: 39.008N

Lon: 74.908W

Elev: 23

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03726

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.7	(c) 19.0	(d) -1.8	(e) 5.0	(f) 17.9	(g) 1.5	(h) 5.9	(i) 21.0	(j) 26.6	(k) 39.9	(l) 23.9	(m) 38.1	(n) 9.8	(o) 310	(p) 0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	90.1	76.6	87.8	75.2	84.3	74.0	78.3	86.4	77.1	84.4	75.8	82.5	9.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	134.0	82.3	74.7	130.2	81.5	73.2	123.5	80.0	41.9	87.0	40.3	84.4	39.3	82.7	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.6	(b) 19.8	(c) 18.0	(e) 9.2	(f) 93.3	(g) 5.0	(h) 2.1	(i) 5.6	(j) 94.9	(k) 2.7	(l) 96.1	(m) -0.1	(n) 97.3	(o) -3.8	(p) 98.8		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			9.2	93.3	5.0	2.1	5.6	94.9	2.7	96.1	-0.1	97.3	-3.8	98.8		
			7.9	80.6	5.2	1.9	4.1	81.9	1.1	83.0	-1.8	84.1	-5.6	85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	35.0	37.1	43.2	52.4	61.4	71.3	76.4	74.7	68.8	58.5	47.9	40.5
	DBStd	16.20	9.73	8.91	8.25	7.91	7.37	5.90	4.99	4.92	6.09	7.70	8.22	8.38
	HDD50	1612	475	370	243	59	2	0	0	0	13	135	315	
	HDD65	4487	931	781	676	387	161	14	0	1	32	231	514	759
	CDD50	3695	9	9	33	133	357	638	818	765	564	276	72	21
	CDD65	1093	0	0	0	10	51	202	353	301	145	30	1	0
	CDH74	7634	0	0	5	74	305	1287	2917	2214	731	100	1	0
	CDH80	2110	0	0	0	18	67	334	966	587	126	12	0	0
	WSAvg	8.7	9.5	9.0	10.2	10.2	8.8	8.1	7.5	7.1	7.7	8.1	8.9	8.9
	PrecAvg	44.9	3.4	2.9	4.3	3.5	3.2	3.7	4.0	4.9	3.5	3.9	3.1	4.4
(15) Precipitation	PrecMax	59.3	6.9	5.9	9.1	6.0	6.4	10.1	7.1	12.3	7.0	8.5	6.5	8.8
	PrecMin	33.2	1.7	0.6	0.4	1.6	0.9	1.5	1.3	2.0	0.4	0.0	0.8	1.7
	PrecStd	5.4	1.3	1.3	2.1	1.2	1.4	2.2	1.4	2.7	1.8	2.1	1.7	1.9
	PrecStd	5.4	1.3	1.3	2.1	1.2	1.4	2.2	1.4	2.7	1.8	2.1	1.7	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	63.1	64.5	71.8	82.2	86.3	90.7	94.5	91.9	88.1	81.3	71.6	65.8	
	0.4% MCWB	55.7	56.4	58.8	65.8	72.1	76.7	79.0	78.2	75.8	70.5	62.7	59.1	
	2% DB	57.3	58.4	64.1	74.6	81.4	87.7	90.5	88.4	82.9	76.7	66.1	61.0	
	2% MCWB	52.9	51.8	54.9	61.6	69.2	74.7	76.6	76.3	73.2	69.1	60.5	56.6	
	5% DB	54.2	54.1	59.5	69.7	77.4	83.7	88.2	85.8	81.0	73.0	63.5	57.0	
	5% MCWB	50.3	48.5	52.0	59.3	66.4	73.1	75.5	75.3	71.9	67.3	58.0	53.2	
	10% DB	49.7	50.0	55.1	64.3	73.1	81.3	85.5	82.7	78.7	70.4	61.1	54.2	
	10% MCWB	46.1	45.6	49.4	56.0	64.2	71.3	74.3	73.5	70.7	64.8	55.9	50.3	
	0.4% WB	57.5	58.4	61.2	67.6	73.9	78.1	81.0	80.2	78.0	73.8	65.1	60.5	
	0.4% MCDB	60.9	62.4	69.6	79.2	82.9	87.8	90.1	89.0	83.7	77.7	68.4	64.5	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	54.1	53.4	56.6	63.6	70.8	76.2	78.8	78.1	75.7	71.1	62.0	57.5	
	2% MCDB	57.1	56.3	62.4	71.8	79.4	84.6	87.3	85.6	81.1	75.0	65.2	60.9	
	5% WB	50.6	49.2	53.4	60.5	68.4	74.7	77.3	76.8	74.2	68.4	59.2	54.2	
	5% MCDB	54.1	52.9	57.2	67.5	75.9	81.9	85.2	83.4	78.9	72.4	62.9	56.8	
	10% WB	46.1	45.7	50.4	57.7	65.9	73.0	75.9	75.4	72.8	65.9	56.8	50.8	
	10% MCDB	48.8	48.9	55.4	63.7	72.2	79.2	83.0	81.4	76.9	69.4	60.8	54.0	
(35) Mean Daily Temperature Range	MDBR	13.2	14.2	14.3	15.7	15.5	14.5	14.9	14.4	14.8	15.3	14.4	13.6	
	5% DB	17.7	19.7	20.5	22.5	20.2	17.3	17.0	16.0	15.6	16.4	16.4	16.4	
	5% WB	15.1	15.2	14.1	12.9	11.1	8.8	7.7	7.3	8.4	10.6	12.7	14.9	
	MCDBR	17.0	18.1	18.0	19.1	18.3	16.0	15.4	13.9	13.1	13.4	14.3	15.7	
	MCWBR	15.8	15.9	14.8	12.7	11.2	8.7	7.6	7.1	7.9	9.6	12.9	15.2	
	MCWBR	15.8	15.9	14.8	12.7	11.2	8.7	7.6	7.1	7.9	9.6	12.9	15.2	
(40) Clear-Sky Solar Irradiance	taub	0.317	0.320	0.361	0.391	0.438	0.491	0.504	0.499	0.417	0.363	0.331	0.317	
	taud	2.514	2.507	2.424	2.364	2.263	2.159	2.136	2.134	2.346	2.498	2.536	2.550	
	Ebn at Noon	273	288	285	281	268	254	249	246	262	268	265	265	
	Edh at Noon	24	28	34	38	43	48	48	47	36	28	24	22	
(44) All-Sky Solar Radiation	RadAvg	650	927	1251	1605	1804	1973	1960	1739	1423	1044	743	572	
	RadStd	57	98	103	121	206	158	96	116	138	103	68	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (61 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+140	

Nomenclature: See separate page